

Lab Session n. 5:  
Solve the Single-Source Warehouse Location Problem  
by Local Search using the Framework EASYLOCAL++

Advanced Scheduling Systems  
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Homework

## Preliminaries

Download the framework EASYLOCAL++ using `git` with the command

```
git clone https://bitbucket.org/satt/easylocal-3
```

Download the source files of Lab Session 4 in one folder. Modify the `Makefile` to point to the actual position of `easylocal-3` in your file system. Compile the code using `make`.

The code considers the Single-Source Warehouse Location Problem. The search space used is the same used by the enumeration algorithm, including the possibility that a warehouse is overloaded. Therefore, there are three cost components: `WL.Capacity` (hard), `WL.Opening` (soft), and `WL.Supply` (soft). Inspect the available code for their implementation.

The move is represented by the change of the supplier for one store and is implemented by the class `WL.Change`, which is also already available.

Finally, the generation of the initial solution is also provided, both using a random procedure and a greedy one.

## Assignment 1

Fill in the missing part of the code, which are:

- All the missing methods of the neighborhood explorer which is represented by the class `WL.ChangeNeighborhoodExplorer`.
- The method `ComputeDeltaCost` of the subclasses of `DeltaCostComponent` for the three cost components.

## Assignment 2

Test the code of Assignment 1 by running the tester of EASYLOCAL++; in particular, verify that the delta costs are correct in various instances and various states.

### Assignment 3

Report the costs obtained by the runner `HillClimbing` on all available instances, with  $10^7$  total evaluations and  $10^6$  iterations from the last improvement. To this aim use the command

```
wlp --main::instance ../Instances/wlpXX.dzn --main::method HC
--HC::max_evaluations 100000000 --HC::max_idle_iterations 1000000
```

### Assignment 4 (optional)

Adapt the script `batch_SA.py` developed for Bus Driver Scheduling to do some (naive) parameter tuning for Simulated Annealing. Report the best parameter configuration and its results.

Send the final version of your work to `andrea.schaerf@uniud.it` (tentative deadline: Sunday, June 14, 2026).